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Web-Assotest: a browser-based tool for genetic association analysis under multiple inheritance models

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Software

- Review
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Summary

`Web-Assotest` is a browser-based tool for analysing various associations between a genetic variant and a binary outcome (cases vs. controls) from a 3×2 table of observed genotype counts (wild-type / heterozygote / homozygote, by case/control status).

The tool simultaneously fits a hierarchy of five nested logistic-regression models — null, dominant, additive (log-additive), recessive, and genotypic — and presents the resulting odds ratios, 95% confidence intervals, and likelihood-ratio test statistics in a model-comparison diagram, with AIC values in an accompanying table, together with a Hardy-Weinberg equilibrium (HWE) check for cases, controls, and the combined sample (Wigginton et al., 2005). Two complementary one-degree-of-freedom tests are reported alongside the model hierarchy: the Cochran–Armitage trend test (Armitage, 1955; Cochran, 1954), which is valid without assuming HWE, and the allelic test with its allelic odds ratio, which doubles the effective sample size but assumes HWE (Sasieni, 1997). Users can optionally fit the same model hierarchy with Firth's penalized logistic regression (Firth, 1993; Ploner et al., 2023) to obtain stable estimates when a genotype count is zero or the data are otherwise separated.

`Web-Assotest` is implemented in R as a single `flexdashboard`/Shiny application (Chang et al., 2024; Iannone et al., 2024; R Core Team, 2024), runs entirely in the browser once deployed, and requires no local software installation or programming beyond entering the genotype counts into an editable table (see Figure 1).

`Web-Assotest` started as a stand-alone Windows programme, `Assotest`, was later ported to the web, and is now released as open-source software with substantially extended functionality. This paper describes version 1.00. The source code is available at https://github.com/ekstroem/WebAssotest, and users who do not want to run the application locally can use it directly in a browser at http://ekstroem.com:3838/webassotest/.

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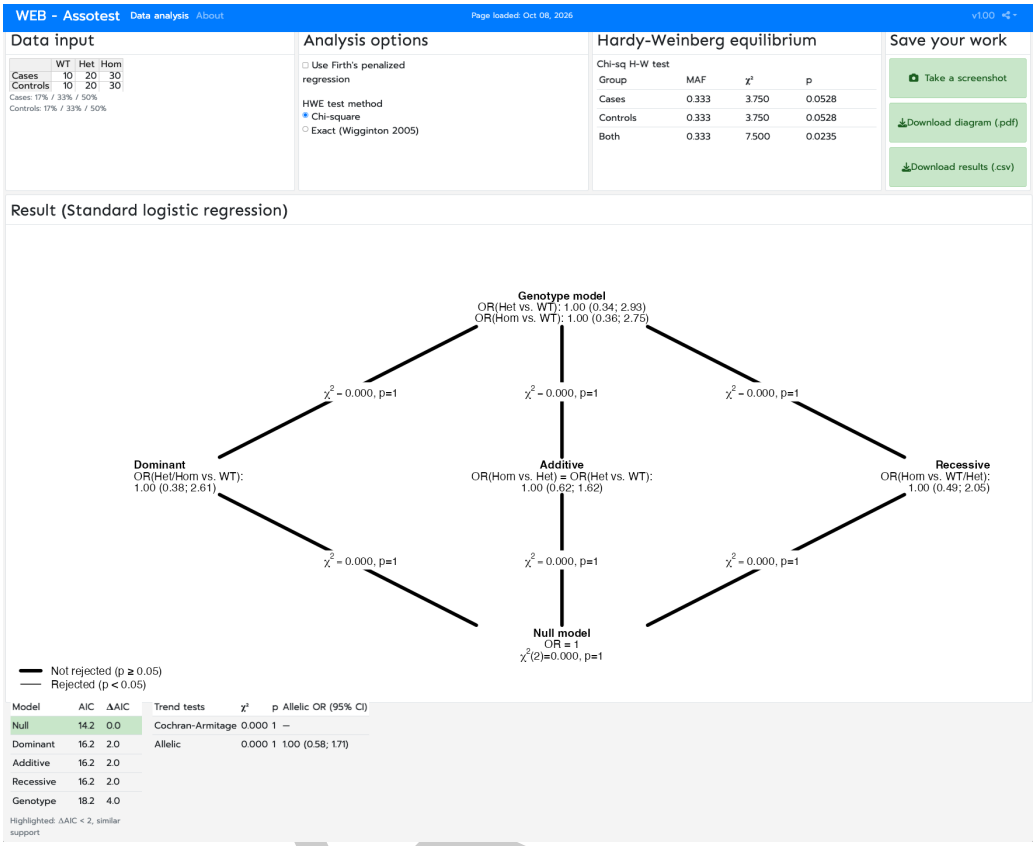


Figure 1: The web-interface for Web-Assotest. Genotype counts and labels are entered in the top-left panel, analysis options are chosen in the next panel, and the HWE results are shown next to them. The model-comparison diagram, the AIC table, and the trend and allelic tests are shown below. Results can be saved as a screenshot, a PDF of the diagram, or a CSV file from the top-right panel.

Statement of need

Testing whether a genetic variant is associated with a disease or trait from a simple genotype-by-outcome count table is one of the most common analyses in applied genetic epidemiology, and it is one that many applied researchers (clinicians, wet-lab biologists, or students) need to perform without necessarily writing R or Python code themselves.

The statistical subtlety is that "the" genetic effect is not a single quantity: the same data can be tested under a dominant, recessive, additive, or fully genotypic model, and these choices are not interchangeable. A variant that shows a strong dominant effect can show a null recessive effect, and vice versa. We use "additive" for the log-additive (per-allele) model and "genotypic" for the two-degrees-of-freedom model, since "co-dominant" is used for either in the literature.

Existing options for this analysis are either large, general-purpose genetics toolchains aimed at genome-wide data (e.g., PLINK (Purcell et al., 2007)) that are overkill and require command-line fluency for a single 3×2 table, or R packages such as `SNPassoc` (González et al., 2007) that require the user to already be working in R. `Web-Assotest` fills the gap between these: a lightweight, model-comprehensive, point-and-click tool for the single-marker case, so that the choice of genetic model is made visible and testable rather than assumed.

The exact HWE test of Wigginton et al. (2005) is bundled so that genotyping-quality checks and the full set of association tests and their extensions are easily available in one place.

The tool has been used as teaching material for case-control study design and has been applied

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by researchers in several independent groups in their applied genetics research (see Research impact statement below).

State of the field

Users who need to analyse a single genotype-by-outcome table have three main kinds of alternatives. Whole-genome toolchains such as PLINK are powerful for genome-wide data, but require installation, input files in specific formats, and command-line fluency, and specification of the different genetic inheritance models, which is a heavy burden for a single marker. R packages for candidate-gene association, such as `SNPassoc`, fit the same family of genetic models and are flexible, but require the user to program in R, which excludes many clinical and laboratory collaborators. Finally, standalone online calculators (e.g., HWE calculators) typically provide either an HWE test or a single allelic or trend test, not a comparison of genetic models. Two web tools come closer. SNPStats (Solé et al., 2006) fits similar genetic models, but requires an uploaded file of individual-level genotype data. GeneRiskCalc (Sudershan et al., 2025) works from genotype counts and reports HWE tests and odds ratios with forest plots under several genetic models, but does not test the models against each other. `Web-Assotest` needs only the summary genotype counts, so it can also re-analyse counts reported in published tables (e.g., for teaching or meta-analysis) without any individual-level data leaving the user, and it formally compares the nested models through likelihood-ratio tests and AIC.

`Web-Assotest` differs from all of the above by combining (a) a no-install, browser-based interface, (b) simultaneous fitting and visual comparison of all standard genetic models rather than a single pre-chosen contrast, together with the trend and allelic tests, and (c) an integrated HWE check, in one small single-file application.

Software design

A browser application rather than an R package. The main design decision was to deliver the analysis as a web application rather than as functions in an R package. Packages such as `SNPassoc` (González et al., 2007) offer scriptability and reproducibility, but exclude the clinicians and laboratory scientists who generate most single-marker data. A browser interface removes the need for installation and programming, at the cost that the analysis is not recorded as code. This cost is addressed by the exports described below, and by keeping the statistical core in plain R functions that can be tested and reused outside the interface.

A single file with few dependencies. `Web-Assotest` is one R Markdown file built with `flexdashboard` and Shiny (Chang et al., 2024; Iannone et al., 2024), with interface and server logic in the same document. Deployment is therefore a single file copy to any Shiny server, and the code is easy to audit, which matters for a tool that must remain available for many years on modest infrastructure. For the same reason, the exact HWE test of Wigginton et al. (2005) and the trend and allelic tests are implemented in base R rather than imported from genetics packages; the only dependencies are `flexdashboard`, `rhandsontable`, `shinyscreenshot`, and `logistf`.

Fitting all models at once. Instead of asking the user to choose a genetic model in advance, every edit to the 3×2 table refits all five nested logistic-regression models, and the likelihood-ratio tests between them are drawn as a diagram. The choice of genetic model is thereby made visible and testable rather than assumed, and the diagram shows which simplifications the data support. In the standard mode the models are fitted to the six grouped counts rather than to individual-level data, so refits are near-instant and the results update as the counts are typed.

Sparse data. Zero cells are common in small candidate-gene studies and make standard odds ratios infinite or undefined. By default the tool uses standard logistic regression, which matches most published analyses, and falls back to Fisher's exact test for the affected contrasts.

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Alternatively, Firth's penalized regression (Firth, 1993; Ploner et al., 2023) can be switched on; it gives finite estimates and profile-likelihood confidence intervals without the fallback, at the cost of slightly shrunken estimates and slower fitting for large samples.

Documentation of results. A known drawback of point-and-click tools is that, unlike a script, they leave no record of how a result was obtained. `Web-Assotest` therefore makes every analysis exportable for subsequent documentation. The CSV file records the entered genotype counts, the settings used (regression method, HWE test, genotype labels, application version, and date), and all estimates and tests: odds ratios with confidence intervals and the likelihood-ratio, trend, allelic, and HWE tests. The model-comparison diagram can be saved as a vector PDF for manuscripts, and a screenshot captures the complete interface. Users can thus archive the analysis with their study data, report it in a publication, and reproduce it later by re-entering the saved counts and settings.

Testing. The statistical core is checked by `tests/check.R`, which compares the application's functions with direct `glm`, `fisher.test`, and `prop.trend.test` results, including zero cells, separation, monomorphic groups, and extremely small p-values. The checks run automatically on every push via GitHub Actions.

Research impact statement

`Web-Assotest` (and its predecessor) has been used in published candidate-gene case-control studies spanning more than two decades (2004–2025) and a broad range of research groups and populations, including Danish, Polish, Iranian, Japanese, Iraqi, and Indian cohorts, and a wide range of phenotypes, for example:

- **Type 2 diabetes and its complications**, including diabetic retinopathy and diabetic foot (Gohari-Lasaki et al., 2020; S. Hansen et al., 2005; S. K. Hansen et al., 2004; Keshavarz et al., 2006; Mrozikiewicz-Rakowska et al., 2017).
- **Obesity, adipokines, and ageing** (Al-Ogaidi et al., 2019; Roszkowska-Gancarz et al., 2012).
- **Cardiovascular disease**, including essential hypertension and acute coronary syndrome (Gutowska et al., 2025; Wrzosek et al., 2015).
- **Psychiatric and behavioural phenotypes**, including completed suicide, opioid dependence, and paraphilic sexual offending (Chojnicka et al., 2012, 2013, 2014; Fudalej et al., 2016; Jakubczyk et al., 2017).
- **Neurodegenerative disease** (late-onset Alzheimer's disease) (Fatemi Langroudi et al., 2023).
- **Autoimmune disease** (Graves' disease) (Kurylowicz et al., 2007).
- **Ophthalmology** (Fuchs endothelial corneal dystrophy) (Ołdak et al., 2015).
- **Gastrointestinal disease** (colonic diverticulosis) (Nehring et al., 2023).
- **Reproductive health** (recurrent miscarriage) (Sudhir et al., 2016).

This breadth shows a recurring need across clinical specialties for a quick, transparent comparison of genetic inheritance models for a single marker by researchers who are not statistical programmers.

AI usage disclosure

AI has been used to correct this manuscript for spelling, and Claude Code was used to assemble the initial list of papers citing `Web-Assotest`; the list was subsequently curated manually. The statistical methodology and the model-comparison design of `webassotest.Rmd` predate AI tools. Claude Code was used to update the `flexdashboard` layout, to speed up the reactive expressions, and to refactor parts of the statistical core (formatting and Fisher's-exact helpers, the structure of the returned results). It was also used to implement the trend and allelic tests,

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the diagram legend, the PDF export, the complete CSV record, numerical fixes (HWE for monomorphic groups), and the `tests/check.R` self-checks. All changes were reviewed by the author and are checked by `tests/check.R`.

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